Model Architecture: gemma-3-12b-it

Generated by sg_modelload

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1 Model Overview

Model name	gemma-3-12b-it
Transformers class	`Gemma3ForConditionalGeneration`
Path	`/models/gemma-3-12b-it`
Total parameters	12,187,325,040 (12.19B)
Weight dtype	bfloat16
Hidden size d	3,840
Decoder layers L	48
Query heads H	16
Key/value heads H_{kv}	8
Head dim d_h	256
MLP width d_{ff}	15,360
Vocabulary V	262,208
Max context length	131,072
Sliding window w	1,024
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	$\text{GELU}_{\text{tanh}}$
Vision encoder: layers	27
hidden size / heads	1152 / 16

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (3840)
V	vocabulary size (262,208)
$H,\ H_{kv},\ d_h$	query heads, key/value heads, size of one head
$W_*,\ b_*$	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
$\text{RMSNorm}(x)$	$\gamma \odot x/\sqrt{\text{mean}(x^2)+\epsilon}$ (learned scale γ)
$\text{LN}(x)$	LayerNorm: $\gamma \odot (x-\mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (gemma-3-12b-it as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$ ($V = 262{,}208$). It is a composition of an embedding E_{mm}, $L = 48$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E_{\text{mm}} \tag{1}$$

Every block maps $\mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ ($d = 3840$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.language_model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** There are 2 block types (same function form within a type, different weights). A type map $c : \{1, \dots, L\} \to \{\mathrm{S}, \mathrm{F}\}$ says which type layer ℓ is: $\mathcal{B}^{(\ell)} = \mathcal{B}_{c(\ell)}(\,\cdot\,; \theta_\ell)$. Here $c = ((\mathrm{S}^5\,\mathrm{F})^8)$, read left to right from layer 1. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\mathrm{F}}^{(48)} \circ \cdots \circ \mathcal{B}_{\mathrm{F}}^{(6)} \circ \mathcal{B}_{\mathrm{S}}^{(5)} \circ \mathcal{B}_{\mathrm{S}}^{(4)} \circ \mathcal{B}_{\mathrm{S}}^{(3)} \circ \mathcal{B}_{\mathrm{S}}^{(2)} \circ \mathcal{B}_{\mathrm{S}}^{(1)} \circ E_{\mathrm{mm}} \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \Big(\big(\mathcal{B}_{\mathrm{F}} \circ \mathcal{B}_{\mathrm{S}}^{\circ 5} \big)^{\circ 8} \Big) \circ E_{\mathrm{mm}} \tag{3}$$

Definition 2.3 (Vision encoder)**.** The vision encoder is itself a composition of 27 blocks acting on a sequence of image patches of width 1152:

$$\mathrm{Enc}_{\mathrm{V}} = \mathrm{Post} \circ \mathcal{B}_{\mathrm{V}}^{\circ 27} \circ \mathrm{PatchEmbed} \tag{4}$$

PatchEmbed turns the raw input into vectors and Post is the final norm / pooling / merging (Table 3).

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\mathrm{S}}$	Sliding-window attention block	40	224,148,992	0–4, 6–10, 12–16, 18–22, 24–28, 30–34, 36–40, 42–46
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	8	224,148,992	5, 11, . . . , 47 (every 6th)
$\mathcal{B}_{\mathrm{V}}$	Vision encoder block	27	15,239,504	0–26

Language-model decoder (48 layers). layers 0–47: $[\mathcal{B}_{\mathrm{S}} \times 5, \mathcal{B}_{\mathrm{F}}]$ repeated 8× (checkpoint indices, i.e. $\ell - 1$).

Vision encoder (27 layers). layers 0–26: $[\mathcal{B}_{\mathrm{V}}]$ repeated 27× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.4 (Token embedding)**.** $E : \{1, \dots, V\}^T \to \mathbb{R}^{T \times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V \times d}$ per token:

$$E(x)_t = W_E[x_t, :] \cdot \sqrt{d}, \qquad t = 1, \dots, T. \tag{5}$$

With images, the multimodal embedding replaces the placeholder-token rows of $E(x)$ by projected encoder outputs:

$$E_{\mathrm{mm}}(x, \cdot) = \mathrm{Merge}\Big(E(x),\ \mathrm{Proj}_{\mathrm{img}}\big(\mathrm{Enc}_{\mathrm{V}}(\mathrm{img}) \big) \Big). \tag{6}$$

Definition 2.5 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\mathrm{final}}$). LN (LayerNorm, used in some encoder blocks) additionally subtracts the mean and adds a bias.

Definition 2.6 (Block $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T \times 3840}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{S}} = (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pf}} \circ \mathrm{FFN} \circ \mathrm{RMSNorm}_{\mathrm{pre}}) \circ (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pa}} \circ \mathrm{Attn} \circ \mathrm{RMSNorm}_{\mathrm{in}}) \tag{7}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention} \\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP} \\
Q &= W_Q\, u && H = 16 \text{ heads of size } d_h = 256 \\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads} \\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}
\end{aligned}$$

$$
\begin{aligned}
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal + sliding window of } w = 1024\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H] &&\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u) \odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width } 15360
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 224{,}148{,}992$): $W_Q \in \mathbb{R}^{4096\times 3840}$, $W_K \in \mathbb{R}^{2048\times 3840}$, $W_V \in \mathbb{R}^{2048\times 3840}$, $W_O \in \mathbb{R}^{3840\times 4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{15360\times 3840}$, $W_{\mathrm{up}} \in \mathbb{R}^{15360\times 3840}$, $W_{\mathrm{down}} \in \mathbb{R}^{3840\times 15360}$, plus 15,872 parameters in norm scales, biases and other vectors. This type is used 40 times.

Definition 2.7 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 3840}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$
\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pf}} \circ \mathrm{FFN} \circ \mathrm{RMSNorm}_{\mathrm{pre}}) \circ (\mathrm{id} + \mathrm{RMSNorm}_{\mathrm{pa}} \circ \mathrm{Attn} \circ \mathrm{RMSNorm}_{\mathrm{in}}) \tag{8}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\,\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\,\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
Q &= W_Q\, u && H = 16 \text{ heads of size } d_h = 256\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H] &&\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u) \odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width } 15360
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 224{,}148{,}992$): $W_Q \in \mathbb{R}^{4096\times 3840}$, $W_K \in \mathbb{R}^{2048\times 3840}$, $W_V \in \mathbb{R}^{2048\times 3840}$, $W_O \in \mathbb{R}^{3840\times 4096}$, $W_{\mathrm{gate}} \in \mathbb{R}^{15360\times 3840}$, $W_{\mathrm{up}} \in \mathbb{R}^{15360\times 3840}$, $W_{\mathrm{down}} \in \mathbb{R}^{3840\times 15360}$, plus 15,872 parameters in norm scales, biases and other vectors. This type is used 8 times.

Definition 2.8 (Block $\mathcal{B}_{\mathrm{V}}$: Vision encoder block)**.** A vision-encoder block of this type maps $h \in \mathbb{R}^{T\times 1152}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$
\mathcal{B}_{\mathrm{V}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{LN}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{LN}_1) \tag{9}
$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u + b_Q && H = 16 \text{ heads of size } d_h = 72\\
K,\ V &= W_K\, u + b_K, \quad W_V\, u + b_V && H_{kv} = 16 \text{ key/value heads}\\
S_h &= s\, Q_h K_h^{\top} + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_h &&\\
\mathrm{Attn}(u) &= W_O\,[O_1; \ldots; O_H] + b_O &&\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width } 4304
\end{aligned}
$$

Parameters of one block ($|\theta_\ell| = 15{,}239{,}504$): $W_K \in \mathbb{R}^{1152\times 1152}$, $W_V \in \mathbb{R}^{1152\times 1152}$, $W_Q \in \mathbb{R}^{1152\times 1152}$, $W_O \in \mathbb{R}^{1152\times 1152}$, $W_1 \in \mathbb{R}^{4304\times 1152}$, $W_2 \in \mathbb{R}^{1152\times 4304}$, plus 14,672 parameters in norm scales, biases and other vectors. This type is used 27 times.

Definition 2.9 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$
\Psi(h)_t = W_E^{\top}\ \mathrm{RMSNorm}_{\mathrm{final}}(h)_t \tag{10}
$$

2.3 Neurons and attention heads

Definition 2.10 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 15360$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{GELU}_{\tanh}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{11}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{12}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 737{,}280$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.11 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{13}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 48 attention layers $\times$ $H = 16$ heads $= 768$ attention heads.

2.4 Architecture and parameters

Definition 2.12 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (262{,}208,\ 3840,\ 48,\ ((\mathrm{S}^5\,\mathrm{F})^8),\ 16,\ 8,\ 256,\ 15360). \tag{14}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}},\ \theta_{\mathrm{Encv}},\ \theta_{\mathrm{Proj}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 40 \times 224{,}148{,}992 + 8 \times 224{,}148{,}992 + 27 \times 15{,}239{,}504 \\ &\quad + 1{,}016{,}706{,}816 \\ &= 12{,}187{,}325{,}040, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, encoder stems and projector, …; Table 3).

2.5 Output representations

Definition 2.13 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions (sliding-window layers additionally hide positions more than w tokens back). Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E_{\text{mm}}}_{F_{\ell-1}\ \text{(prefix)}} \tag{15}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E_{\text{mm}}$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E_{\text{mm}}(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \text{RMSNorm}_{\text{pa}}\big(\text{Attn}_\ell(\text{RMSNorm}_{\text{in}}(h^{(\ell-1)}))\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \text{RMSNorm}_{\text{pre}}(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \text{RMSNorm}_{\text{pf}}\big(\text{FFN}_\ell(u^{(\ell)})\big)$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{16}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, $\text{GELU}_{\text{tanh}}$ and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.14 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\text{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{17}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{18}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E_{\text{mm}},\ \Psi$	embedding and output head
$\text{Enc}_{\text{V}},\ \text{Enc}_{\text{A}},\ \text{Proj}$	vision / audio encoders and their projectors into the text width
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\text{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ

continued on next page

Symbol	Meaning
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 gemma-3-12b-it: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow E(x) \cdot \sqrt{d}$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}3840$
if image input present **then**
  $e \leftarrow \mathrm{Enc_V}(\text{image})$ ▷ vision encoder: 27 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 3840
end if
$\ell \leftarrow 0$
for $r \leftarrow 1$ **to** 8 **do** ▷ layers 1–48: pattern of 6 blocks, repeated 8 times
  for $j \leftarrow 1$ **to** 5 **do**
    $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_\mathrm{S}(h; \theta_\ell)$ ▷ Sliding-window attention block
  end for
  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_\mathrm{F}(h; \theta_\ell)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm_{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ
$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}262,208$, embedding matrix re-used
return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 1$ **to** 27 **do** ▷ 27 blocks of type $\mathcal{B}_\mathrm{V}$
  $h \leftarrow \mathcal{B}_\mathrm{V}(h; \theta_\ell)$ ▷ Vision encoder block; result is $h^{(\ell)}$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 3
return h

Algorithm 3 Block $\mathcal{B}_\mathrm{S}(h)$ — Sliding-window attention block

Require: h of shape $B \times T \times 3840$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 key/value heads, each shared by 2 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal + only the last 1024 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}3840$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU_{tanh}}(uW_{\text{gate}}^\top) \odot uW_{\text{up}}^\top\big) W_{\text{down}}^\top$ ▷ expand to 15360, gate, shrink back to 3840
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 3840$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 key/value heads, each shared by 2 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}3840$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\mathrm{tanh}}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 15360, gate, shrink back to 3840
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 1152$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}16{\times}72$ (16 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}16{\times}72$ (16 key/value heads)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}72$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}1152$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\mathrm{tanh}}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 4304, shrink back to 1152
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.vision_tower.embeddings.patch_embedding.weight`	Conv2d	$1152 \times 3 \times 14 \times 14$	677,376
`model.vision_tower.embeddings.patch_embedding.bias`	Conv2d	1152	1,152
`model.vision_tower.embeddings.position_embedding.weight`	Embedding	4096×1152	4,718,592
`model.vision_tower.post_layernorm.weight`	LayerNorm	1152	1,152
`model.vision_tower.post_layernorm.bias`	LayerNorm	1152	1,152
`model.multi_modal_projector.mm_input_projection_weight`	Gemma3MultiModalProjector	1152×3840	4,423,680
`model.multi_modal_projector.mm_soft_emb_norm.weight`	Gemma3RMSNorm	1152	1,152

continued on next page

Weight	Type	Shape	Params
`model.language_model.embed_tokens.weight`	Gemma3TextScaledWordEmbedding	262208×3840	1,006,878,720
`model.language_model.norm.weight`	Gemma3RMSNorm	3840	3,840
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	262208×3840	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 4: $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block — layers 0–4, 6–10, 12–16, 18–22, 24–28, 30–34, 36–40, 42–46; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×3840	15,728,640
`self_attn.k_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.v_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.o_proj.weight`	Linear	3840×4096	15,728,640
`self_attn.q_norm.weight`	Gemma3RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma3RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	15360×3840	58,982,400
`mlp.up_proj.weight`	Linear	15360×3840	58,982,400
`mlp.down_proj.weight`	Linear	3840×15360	58,982,400
`input_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_attention_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
Per block			**224,148,992**
× 40 blocks			**8,965,959,680**

Table 5: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 5, 11, …, 47 (every 6th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×3840	15,728,640
`self_attn.k_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.v_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.o_proj.weight`	Linear	3840×4096	15,728,640
`self_attn.q_norm.weight`	Gemma3RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma3RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	15360×3840	58,982,400
`mlp.up_proj.weight`	Linear	15360×3840	58,982,400
`mlp.down_proj.weight`	Linear	3840×15360	58,982,400
`input_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_attention_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
Per block			**224,148,992**
× 8 blocks			**1,793,191,936**

4.3 Vision encoder: `model.vision_tower.encoder.layers`

Table 6: $\mathcal{B}_V$: Vision encoder block — layers 0–26; weight names are relative to `model.vision_tower.encoder.layers.`ℓ

Weight (relative)	Type	Shape	Params
`layer_norm1.weight`	LayerNorm	1152	1,152
`layer_norm1.bias`	LayerNorm	1152	1,152
`self_attn.k_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.k_proj.bias`	Linear	1152	1,152
`self_attn.v_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.v_proj.bias`	Linear	1152	1,152
`self_attn.q_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.q_proj.bias`	Linear	1152	1,152
`self_attn.out_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.out_proj.bias`	Linear	1152	1,152
`layer_norm2.weight`	LayerNorm	1152	1,152
`layer_norm2.bias`	LayerNorm	1152	1,152
`mlp.fc1.weight`	Linear	4304×1152	4,958,208
`mlp.fc1.bias`	Linear	4304	4,304
`mlp.fc2.weight`	Linear	1152×4304	4,958,208
`mlp.fc2.bias`	Linear	1152	1,152
Per block			**15,239,504**
× 27 blocks			**411,466,608**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).

continued on next page

Layer (appears as)	What it does
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
Patch embedding (Conv2d) `embeddings.patch_embedding`	A strided convolution that cuts the image into square patches (14×14 pixels) and maps each patch to one vector.
Position embedding `embeddings.position_embedding`	One learned vector per patch position, added so the encoder knows where each patch is.
Pre-attention LayerNorm `layers.layer_norm1`	LayerNorm (mean/variance normalisation with scale and bias) before self-attention.
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Attention output projection `self_attn.out_proj`	Mixes the concatenated head outputs back to the model width.
Pre-MLP LayerNorm `layers.layer_norm2`	LayerNorm before the MLP.
MLP up-projection (fc1) `mlp.fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.fc2`	Shrinks back to the model width.
Final vision LayerNorm `vision_tower.post_layernorm`	Normalises the vision encoder's output patches.
MULTIMODAL PROJECTOR	
Image-to-text projection `multi_modal_projector`	Matrix mapping each pooled image feature (vision width) into the language model's hidden size.
Projector norm `multi_modal_projector.mm_soft_emb_norm`	RMSNorm on image features after average-pooling the 4096 patches down to 256 image tokens.

6 Parameter Count

Table 8: Parameters by component

Component	Parameters	% of total
MLP / MoE	8,493,465,600	69.7%
Attention / token mixer	2,264,948,736	18.6%
Token embedding	1,006,878,720	8.3%
Vision encoder	416,866,032	3.4%
Multimodal projector	4,424,832	0.0%
Norms	741,120	0.0%
Total	**12,187,325,040**	**100.0%**

Table 9: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{S}}$ Sliding-window attention block	40	224,148,992	8,965,959,680
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	8	224,148,992	1,793,191,936
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	27	15,239,504	411,466,608
Modules outside stacks	1	–	1,016,706,816
Total			**12,187,325,040**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	262208 × 3840	bfloat16	1,006,878,720
`model`	Gemma3Model	–	–	–
`model.language_model`	Gemma3TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Gemma3TextScaled WordEmbedding	262208 × 3840	bfloat16	1,006,878,720
`model.language_model.layers.0`	Gemma3DecoderLayer	–	–	–
`model.language_model.layers.0. input_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0.mlp`	Gemma3MLP	–	–	–
`model.language_model.layers.0.mlp.down_proj. weight`	Linear	3840 × 15360	bfloat16	58,982,400
`model.language_model.layers.0.mlp.gate_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.0.mlp.up_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.0. post_attention_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0. post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0. pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0.self_attn`	Gemma3Attention	–	–	–
`model.language_model.layers.0.self_attn. k_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.0.self_attn. k_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320
`model.language_model.layers.0.self_attn. o_proj.weight`	Linear	3840 × 4096	bfloat16	15,728,640
`model.language_model.layers.0.self_attn. q_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.0.self_attn. q_proj.weight`	Linear	4096 × 3840	bfloat16	15,728,640
`model.language_model.layers.0.self_attn. v_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320
`model.language_model.layers.1`	Gemma3DecoderLayer	–	–	–
`model.language_model.layers.1. input_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1.mlp`	Gemma3MLP	–	–	–
`model.language_model.layers.1.mlp.down_proj. weight`	Linear	3840 × 15360	bfloat16	58,982,400
`model.language_model.layers.1.mlp.gate_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.1.mlp.up_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.1. post_attention_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1. post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1. pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1.self_attn`	Gemma3Attention	–	–	–
`model.language_model.layers.1.self_attn. k_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.1.self_attn. k_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.1.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.1.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.1.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.2	Gemma3DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.mlp	Gemma3MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.2.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.2.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.2.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.2.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.2.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.2.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.2.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.3	Gemma3DecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.mlp	Gemma3MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.3.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.3.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.3.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.3.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.3.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.3.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.4	Gemma3DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.mlp	Gemma3MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.4.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.4.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.4.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.4.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.4.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.4.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.4.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.4.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.4.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.5	Gemma3DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.mlp	Gemma3MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.5.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.5.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.5.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.5.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.5.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.5.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.5.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.6	Gemma3DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.mlp	Gemma3MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.6.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.6.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.6.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.6.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.6.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.6.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.6.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.7	Gemma3DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.mlp	Gemma3MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.7.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.7.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.7.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.7.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.7.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.7.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.8	Gemma3DecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.mlp	Gemma3MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.8.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.8.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.8.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.8.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.8.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.8.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.8.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.9	Gemma3DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.mlp	Gemma3MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.9.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.9.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.9.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.9.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.9.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.9.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.9.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.9.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.9.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.10	Gemma3DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.mlp	Gemma3MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.10.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.10.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.10.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.10.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.10.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.10.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.10.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.11	Gemma3DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.mlp	Gemma3MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.11.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.11.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.11.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.11.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.11.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.11.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.11.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.12	Gemma3DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.mlp	Gemma3MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.12.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.12.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.12.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.12.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.12.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.12.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.12.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.13	Gemma3DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.mlp	Gemma3MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.13.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.13.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.13.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.13.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.13.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.13.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.13.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.14	Gemma3DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.mlp	Gemma3MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.14.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.14.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.14.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.14.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.14.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.14.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.14.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.14.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.14.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.15	Gemma3DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.mlp	Gemma3MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.15.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.15.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.15.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.15.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.15.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.15.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.16	Gemma3DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.mlp	Gemma3MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.16.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.16.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.16.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.16.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.16.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.16.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.16.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.17	Gemma3DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.mlp	Gemma3MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.17.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.17.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.17.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.17.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.17.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.17.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.17.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.18	Gemma3DecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.mlp	Gemma3MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.18.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.18.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.18.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.18.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.18.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.18.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.18.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.19	Gemma3DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.mlp	Gemma3MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.19.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.19.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.19.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.19.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.19.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.19.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.20	Gemma3DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.mlp	Gemma3MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.20.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.20.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.20.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.20.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.20.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.20.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.20.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.21	Gemma3DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.mlp	Gemma3MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.21.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.21.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.21.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.21.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.21.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.21.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.21.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.22	Gemma3DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.mlp	Gemma3MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.22.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.22.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.22.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.22.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.22.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.22.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.22.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.23	Gemma3DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.mlp	Gemma3MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.23.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.23.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.23.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.23.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.23.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.23.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.24	Gemma3DecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.mlp	Gemma3MLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.24.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.24.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.24.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.self_attn	Gemma3Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.24.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.24.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.24.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.24.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.24.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.24.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.25	Gemma3DecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.mlp	Gemma3MLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.25.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.25.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.25.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.25.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.25.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.25.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.25.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.26	Gemma3DecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.mlp	Gemma3MLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.26.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.26.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.26.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.26.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.26.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.26.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.26.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.27	Gemma3DecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.mlp	Gemma3MLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.27.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.27.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.27.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.27.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.27.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.27.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.27.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.27.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.28	Gemma3DecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.mlp	Gemma3MLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.28.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.28.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.28.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.28.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.28.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.28.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.28.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.29	Gemma3DecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.mlp	Gemma3MLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.29.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.29.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.29.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.29.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.29.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.29.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.29.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.29.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.29.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.30	Gemma3DecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.mlp	Gemma3MLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.30.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.30.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.30.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.30.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.30.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.30.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.30.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.31	Gemma3DecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.mlp	Gemma3MLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.31.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.31.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.31.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.31.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.31.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.31.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.31.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.32	Gemma3DecoderLayer	–	–	–
model.language_model.layers.32.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.mlp	Gemma3MLP	–	–	–
model.language_model.layers.32.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.32.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.32.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.32.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.32.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.32.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.32.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.32.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.33	Gemma3DecoderLayer	–	–	–
model.language_model.layers.33.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.mlp	Gemma3MLP	–	–	–
model.language_model.layers.33.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.33.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.33.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.33.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.33.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.33.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.33.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.33.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.34	Gemma3DecoderLayer	–	–	–
model.language_model.layers.34.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.mlp	Gemma3MLP	–	–	–
model.language_model.layers.34.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.34.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.34.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.34.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.34.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.34.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.34.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.34.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.34.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.34.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.35	Gemma3DecoderLayer	–	–	–
model.language_model.layers.35.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.mlp	Gemma3MLP	–	–	–
model.language_model.layers.35.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.35.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.35.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.35.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.35.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.35.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.35.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.35.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.35.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.35.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.36	Gemma3DecoderLayer	–	–	–
model.language_model.layers.36.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.mlp	Gemma3MLP	–	–	–
model.language_model.layers.36.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.36.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.36.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.36.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.36.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.36.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.36.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.36.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.36.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.36.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.37	Gemma3DecoderLayer	–	–	–
model.language_model.layers.37.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.mlp	Gemma3MLP	–	–	–
model.language_model.layers.37.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.37.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.37.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.37.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.37.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.37.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.37.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.37.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.37.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.37.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.38	Gemma3DecoderLayer	–	–	–
model.language_model.layers.38.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.mlp	Gemma3MLP	–	–	–
model.language_model.layers.38.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.38.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.38.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.38.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.38.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.38.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.38.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.38.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.38.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.38.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.39	Gemma3DecoderLayer	–	–	–
model.language_model.layers.39.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.mlp	Gemma3MLP	–	–	–
model.language_model.layers.39.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.39.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.39.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.39.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.39.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.39.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.39.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.39.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.39.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.39.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.40	Gemma3DecoderLayer	–	–	–
model.language_model.layers.40.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.mlp	Gemma3MLP	–	–	–
model.language_model.layers.40.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.40.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.40.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.40.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.40.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.40.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.40.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.40.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.40.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.40.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.41	Gemma3DecoderLayer	–	–	–
model.language_model.layers.41.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.mlp	Gemma3MLP	–	–	–
model.language_model.layers.41.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.41.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.41.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.41.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.41.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.41.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.41.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.41.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.41.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.41.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.42	Gemma3DecoderLayer	–	–	–
model.language_model.layers.42.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.mlp	Gemma3MLP	–	–	–
model.language_model.layers.42.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.42.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.42.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.42.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.42.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.42.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.42.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.42.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.42.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.42.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.43	Gemma3DecoderLayer	–	–	–
model.language_model.layers.43.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.mlp	Gemma3MLP	–	–	–
model.language_model.layers.43.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.43.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.43.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.43.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.43.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.43.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.43.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.43.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.43.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.43.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.44	Gemma3DecoderLayer	–	–	–
model.language_model.layers.44.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.mlp	Gemma3MLP	–	–	–
model.language_model.layers.44.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.44.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.44.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.44.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.44.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.44.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.44.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.44.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.44.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.44.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.45	Gemma3DecoderLayer	–	–	–
model.language_model.layers.45.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.mlp	Gemma3MLP	–	–	–
model.language_model.layers.45.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.45.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.45.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.45.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.45.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.45.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.45.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.45.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.45.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.45.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.46	Gemma3DecoderLayer	–	–	–
model.language_model.layers.46.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.mlp	Gemma3MLP	–	–	–
model.language_model.layers.46.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.46.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.46.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.46.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.46.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.46.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.46.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.46.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.46.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.46.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.47	Gemma3DecoderLayer	–	–	–
model.language_model.layers.47.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.mlp	Gemma3MLP	–	–	–
model.language_model.layers.47.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.47.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.47.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.47.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.47.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.47.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.47.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.47.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.47.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.47.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.47.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.norm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.multi_modal_projector.mm_input_projection_weight	Gemma3MultiModalProjector	1152 × 3840	bfloat16	4,423,680
model.multi_modal_projector	Gemma3MultiModalProjector	–	–	–
model.multi_modal_projector.mm_soft_emb_norm.weight	Gemma3RMSNorm	1152	bfloat16	1,152
model.vision_tower	SiglipVisionModel	–	–	–
model.vision_tower.embeddings	SiglipVisionEmbeddings	–	–	–
model.vision_tower.embeddings.patch_embedding.bias	Conv2d	1152	bfloat16	1,152
model.vision_tower.embeddings.patch_embedding.weight	Conv2d	1152 × 3 × 14 × 14	bfloat16	677,376
model.vision_tower.embeddings.position_embedding.weight	Embedding	4096 × 1152	bfloat16	4,718,592
model.vision_tower.encoder	SiglipEncoder	–	–	–
model.vision_tower.encoder.layers.0	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.0.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.0.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.0.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.0.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.0.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.1.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.1.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.1.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.1.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.1.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.1.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.2.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.2.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.2.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.2.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.2.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.3.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.3.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.3.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.3.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.3.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.3.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.4.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.4.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.4.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.4.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.4.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.5.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.5.mlp.fc1.bias	Linear	4304	bfloat16	4,304

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.5.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.5.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.5.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.6.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.6.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.6.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.6.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.6.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.7.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.7.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.7.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.7.mlp.fc2.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.7.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.7.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.8.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.8.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.8.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.8.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.8.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.9.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.9.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.9.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.9.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.9.self_attn	SiglipAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.9.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.10.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.10.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.10.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.10.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.10.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.11.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.11.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.11.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.11.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.11.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.11.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.12.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.12.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.12.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.12.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.12.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.13.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.13.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.13.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.13.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.13.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.13.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.14.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.14.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.14.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.14.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.14.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.15.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.15.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.15.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.15.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.15.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.15.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.16.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.16.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.16.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.16.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.16.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.16.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.17.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.17.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.17.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.17.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.17.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.17.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.17.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.18.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.18.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.18.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.18.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.18.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.18.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.19.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.19.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.19.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.19.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.19.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.19.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.19.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.20.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.20.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.20.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.20.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.20.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.20.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.21.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.21.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.21.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.21.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.21.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.21.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.21.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.22.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.22.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.22.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.22.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.22.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.22.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.23.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.23.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.23.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.23.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.23.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.23.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.23.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.24.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.24.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.24.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.24.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.24.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.24.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.25.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.25.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.25.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.25.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.25.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.25.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.26	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.26.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.26.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.26.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.26.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.26.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.26.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.post_layernorm.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.post_layernorm.weight	LayerNorm	1152	bfloat16	1,152

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.language_model.embed_tokens	1,006,878,720	1,006,878,720
model.language_model.layers.0.input_layernorm	3,840	1,006,882,560
model.language_model.layers.0.mlp.down_proj	58,982,400	1,065,864,960
model.language_model.layers.0.mlp.gate_proj	58,982,400	1,124,847,360
model.language_model.layers.0.mlp.up_proj	58,982,400	1,183,829,760
model.language_model.layers.0.post_attention_layernorm	3,840	1,183,833,600
model.language_model.layers.0.post_feedforward_layernorm	3,840	1,183,837,440
model.language_model.layers.0.pre_feedforward_layernorm	3,840	1,183,841,280
model.language_model.layers.0.self_attn.k_norm	256	1,183,841,536
model.language_model.layers.0.self_attn.k_proj	7,864,320	1,191,705,856
model.language_model.layers.0.self_attn.o_proj	15,728,640	1,207,434,496
model.language_model.layers.0.self_attn.q_norm	256	1,207,434,752
model.language_model.layers.0.self_attn.q_proj	15,728,640	1,223,163,392
model.language_model.layers.0.self_attn.v_proj	7,864,320	1,231,027,712
model.language_model.layers.1.input_layernorm	3,840	1,231,031,552
model.language_model.layers.1.mlp.down_proj	58,982,400	1,290,013,952
model.language_model.layers.1.mlp.gate_proj	58,982,400	1,348,996,352
model.language_model.layers.1.mlp.up_proj	58,982,400	1,407,978,752
model.language_model.layers.1.post_attention_layernorm	3,840	1,407,982,592

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Module	Params	Running total
model.language_model.layers.1. post_feedforward_layernorm	3,840	1,407,986,432
model.language_model.layers.1. pre_feedforward_layernorm	3,840	1,407,990,272
model.language_model.layers.1.self_attn.k_norm	256	1,407,990,528
model.language_model.layers.1.self_attn.k_proj	7,864,320	1,415,854,848
model.language_model.layers.1.self_attn.o_proj	15,728,640	1,431,583,488
model.language_model.layers.1.self_attn.q_norm	256	1,431,583,744
model.language_model.layers.1.self_attn.q_proj	15,728,640	1,447,312,384
model.language_model.layers.1.self_attn.v_proj	7,864,320	1,455,176,704
model.language_model.layers.2.input_layernorm	3,840	1,455,180,544
model.language_model.layers.2.mlp.down_proj	58,982,400	1,514,162,944
model.language_model.layers.2.mlp.gate_proj	58,982,400	1,573,145,344
model.language_model.layers.2.mlp.up_proj	58,982,400	1,632,127,744
model.language_model.layers.2. post_attention_layernorm	3,840	1,632,131,584
model.language_model.layers.2. post_feedforward_layernorm	3,840	1,632,135,424
model.language_model.layers.2. pre_feedforward_layernorm	3,840	1,632,139,264
model.language_model.layers.2.self_attn.k_norm	256	1,632,139,520
model.language_model.layers.2.self_attn.k_proj	7,864,320	1,640,003,840
model.language_model.layers.2.self_attn.o_proj	15,728,640	1,655,732,480
model.language_model.layers.2.self_attn.q_norm	256	1,655,732,736
model.language_model.layers.2.self_attn.q_proj	15,728,640	1,671,461,376
model.language_model.layers.2.self_attn.v_proj	7,864,320	1,679,325,696
model.language_model.layers.3.input_layernorm	3,840	1,679,329,536
model.language_model.layers.3.mlp.down_proj	58,982,400	1,738,311,936
model.language_model.layers.3.mlp.gate_proj	58,982,400	1,797,294,336
model.language_model.layers.3.mlp.up_proj	58,982,400	1,856,276,736
model.language_model.layers.3. post_attention_layernorm	3,840	1,856,280,576
model.language_model.layers.3. post_feedforward_layernorm	3,840	1,856,284,416
model.language_model.layers.3. pre_feedforward_layernorm	3,840	1,856,288,256
model.language_model.layers.3.self_attn.k_norm	256	1,856,288,512
model.language_model.layers.3.self_attn.k_proj	7,864,320	1,864,152,832
model.language_model.layers.3.self_attn.o_proj	15,728,640	1,879,881,472
model.language_model.layers.3.self_attn.q_norm	256	1,879,881,728
model.language_model.layers.3.self_attn.q_proj	15,728,640	1,895,610,368
model.language_model.layers.3.self_attn.v_proj	7,864,320	1,903,474,688
model.language_model.layers.4.input_layernorm	3,840	1,903,478,528
model.language_model.layers.4.mlp.down_proj	58,982,400	1,962,460,928
model.language_model.layers.4.mlp.gate_proj	58,982,400	2,021,443,328
model.language_model.layers.4.mlp.up_proj	58,982,400	2,080,425,728
model.language_model.layers.4. post_attention_layernorm	3,840	2,080,429,568
model.language_model.layers.4. post_feedforward_layernorm	3,840	2,080,433,408
model.language_model.layers.4. pre_feedforward_layernorm	3,840	2,080,437,248
model.language_model.layers.4.self_attn.k_norm	256	2,080,437,504
model.language_model.layers.4.self_attn.k_proj	7,864,320	2,088,301,824
model.language_model.layers.4.self_attn.o_proj	15,728,640	2,104,030,464
model.language_model.layers.4.self_attn.q_norm	256	2,104,030,720
model.language_model.layers.4.self_attn.q_proj	15,728,640	2,119,759,360
model.language_model.layers.4.self_attn.v_proj	7,864,320	2,127,623,680

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Module	Params	Running total
model.language_model.layers.5.input_layernorm	3,840	2,127,627,520
model.language_model.layers.5.mlp.down_proj	58,982,400	2,186,609,920
model.language_model.layers.5.mlp.gate_proj	58,982,400	2,245,592,320
model.language_model.layers.5.mlp.up_proj	58,982,400	2,304,574,720
model.language_model.layers.5.post_attention_layernorm	3,840	2,304,578,560
model.language_model.layers.5.post_feedforward_layernorm	3,840	2,304,582,400
model.language_model.layers.5.pre_feedforward_layernorm	3,840	2,304,586,240
model.language_model.layers.5.self_attn.k_norm	256	2,304,586,496
model.language_model.layers.5.self_attn.k_proj	7,864,320	2,312,450,816
model.language_model.layers.5.self_attn.o_proj	15,728,640	2,328,179,456
model.language_model.layers.5.self_attn.q_norm	256	2,328,179,712
model.language_model.layers.5.self_attn.q_proj	15,728,640	2,343,908,352
model.language_model.layers.5.self_attn.v_proj	7,864,320	2,351,772,672
model.language_model.layers.6.input_layernorm	3,840	2,351,776,512
model.language_model.layers.6.mlp.down_proj	58,982,400	2,410,758,912
model.language_model.layers.6.mlp.gate_proj	58,982,400	2,469,741,312
model.language_model.layers.6.mlp.up_proj	58,982,400	2,528,723,712
model.language_model.layers.6.post_attention_layernorm	3,840	2,528,727,552
model.language_model.layers.6.post_feedforward_layernorm	3,840	2,528,731,392
model.language_model.layers.6.pre_feedforward_layernorm	3,840	2,528,735,232
model.language_model.layers.6.self_attn.k_norm	256	2,528,735,488
model.language_model.layers.6.self_attn.k_proj	7,864,320	2,536,599,808
model.language_model.layers.6.self_attn.o_proj	15,728,640	2,552,328,448
model.language_model.layers.6.self_attn.q_norm	256	2,552,328,704
model.language_model.layers.6.self_attn.q_proj	15,728,640	2,568,057,344
model.language_model.layers.6.self_attn.v_proj	7,864,320	2,575,921,664
model.language_model.layers.7.input_layernorm	3,840	2,575,925,504
model.language_model.layers.7.mlp.down_proj	58,982,400	2,634,907,904
model.language_model.layers.7.mlp.gate_proj	58,982,400	2,693,890,304
model.language_model.layers.7.mlp.up_proj	58,982,400	2,752,872,704
model.language_model.layers.7.post_attention_layernorm	3,840	2,752,876,544
model.language_model.layers.7.post_feedforward_layernorm	3,840	2,752,880,384
model.language_model.layers.7.pre_feedforward_layernorm	3,840	2,752,884,224
model.language_model.layers.7.self_attn.k_norm	256	2,752,884,480
model.language_model.layers.7.self_attn.k_proj	7,864,320	2,760,748,800
model.language_model.layers.7.self_attn.o_proj	15,728,640	2,776,477,440
model.language_model.layers.7.self_attn.q_norm	256	2,776,477,696
model.language_model.layers.7.self_attn.q_proj	15,728,640	2,792,206,336
model.language_model.layers.7.self_attn.v_proj	7,864,320	2,800,070,656
model.language_model.layers.8.input_layernorm	3,840	2,800,074,496
model.language_model.layers.8.mlp.down_proj	58,982,400	2,859,056,896
model.language_model.layers.8.mlp.gate_proj	58,982,400	2,918,039,296
model.language_model.layers.8.mlp.up_proj	58,982,400	2,977,021,696
model.language_model.layers.8.post_attention_layernorm	3,840	2,977,025,536
model.language_model.layers.8.post_feedforward_layernorm	3,840	2,977,029,376
model.language_model.layers.8.pre_feedforward_layernorm	3,840	2,977,033,216

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Module	Params	Running total
model.language_model.layers.8.self_attn.k_norm	256	2,977,033,472
model.language_model.layers.8.self_attn.k_proj	7,864,320	2,984,897,792
model.language_model.layers.8.self_attn.o_proj	15,728,640	3,000,626,432
model.language_model.layers.8.self_attn.q_norm	256	3,000,626,688
model.language_model.layers.8.self_attn.q_proj	15,728,640	3,016,355,328
model.language_model.layers.8.self_attn.v_proj	7,864,320	3,024,219,648
model.language_model.layers.9.input_layernorm	3,840	3,024,223,488
model.language_model.layers.9.mlp.down_proj	58,982,400	3,083,205,888
model.language_model.layers.9.mlp.gate_proj	58,982,400	3,142,188,288
model.language_model.layers.9.mlp.up_proj	58,982,400	3,201,170,688
model.language_model.layers.9. post_attention_layernorm	3,840	3,201,174,528
model.language_model.layers.9. post_feedforward_layernorm	3,840	3,201,178,368
model.language_model.layers.9. pre_feedforward_layernorm	3,840	3,201,182,208
model.language_model.layers.9.self_attn.k_norm	256	3,201,182,464
model.language_model.layers.9.self_attn.k_proj	7,864,320	3,209,046,784
model.language_model.layers.9.self_attn.o_proj	15,728,640	3,224,775,424
model.language_model.layers.9.self_attn.q_norm	256	3,224,775,680
model.language_model.layers.9.self_attn.q_proj	15,728,640	3,240,504,320
model.language_model.layers.9.self_attn.v_proj	7,864,320	3,248,368,640
model.language_model.layers.10.input_layernorm	3,840	3,248,372,480
model.language_model.layers.10.mlp.down_proj	58,982,400	3,307,354,880
model.language_model.layers.10.mlp.gate_proj	58,982,400	3,366,337,280
model.language_model.layers.10.mlp.up_proj	58,982,400	3,425,319,680
model.language_model.layers.10. post_attention_layernorm	3,840	3,425,323,520
model.language_model.layers.10. post_feedforward_layernorm	3,840	3,425,327,360
model.language_model.layers.10. pre_feedforward_layernorm	3,840	3,425,331,200
model.language_model.layers.10.self_attn.k_norm	256	3,425,331,456
model.language_model.layers.10.self_attn.k_proj	7,864,320	3,433,195,776
model.language_model.layers.10.self_attn.o_proj	15,728,640	3,448,924,416
model.language_model.layers.10.self_attn.q_norm	256	3,448,924,672
model.language_model.layers.10.self_attn.q_proj	15,728,640	3,464,653,312
model.language_model.layers.10.self_attn.v_proj	7,864,320	3,472,517,632
model.language_model.layers.11.input_layernorm	3,840	3,472,521,472
model.language_model.layers.11.mlp.down_proj	58,982,400	3,531,503,872
model.language_model.layers.11.mlp.gate_proj	58,982,400	3,590,486,272
model.language_model.layers.11.mlp.up_proj	58,982,400	3,649,468,672
model.language_model.layers.11. post_attention_layernorm	3,840	3,649,472,512
model.language_model.layers.11. post_feedforward_layernorm	3,840	3,649,476,352
model.language_model.layers.11. pre_feedforward_layernorm	3,840	3,649,480,192
model.language_model.layers.11.self_attn.k_norm	256	3,649,480,448
model.language_model.layers.11.self_attn.k_proj	7,864,320	3,657,344,768
model.language_model.layers.11.self_attn.o_proj	15,728,640	3,673,073,408
model.language_model.layers.11.self_attn.q_norm	256	3,673,073,664
model.language_model.layers.11.self_attn.q_proj	15,728,640	3,688,802,304
model.language_model.layers.11.self_attn.v_proj	7,864,320	3,696,666,624
model.language_model.layers.12.input_layernorm	3,840	3,696,670,464
model.language_model.layers.12.mlp.down_proj	58,982,400	3,755,652,864
model.language_model.layers.12.mlp.gate_proj	58,982,400	3,814,635,264
model.language_model.layers.12.mlp.up_proj	58,982,400	3,873,617,664

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Module	Params	Running total
model.language_model.layers.12.post_attention_layernorm	3,840	3,873,621,504
model.language_model.layers.12.post_feedforward_layernorm	3,840	3,873,625,344
model.language_model.layers.12.pre_feedforward_layernorm	3,840	3,873,629,184
model.language_model.layers.12.self_attn.k_norm	256	3,873,629,440
model.language_model.layers.12.self_attn.k_proj	7,864,320	3,881,493,760
model.language_model.layers.12.self_attn.o_proj	15,728,640	3,897,222,400
model.language_model.layers.12.self_attn.q_norm	256	3,897,222,656
model.language_model.layers.12.self_attn.q_proj	15,728,640	3,912,951,296
model.language_model.layers.12.self_attn.v_proj	7,864,320	3,920,815,616
model.language_model.layers.13.input_layernorm	3,840	3,920,819,456
model.language_model.layers.13.mlp.down_proj	58,982,400	3,979,801,856
model.language_model.layers.13.mlp.gate_proj	58,982,400	4,038,784,256
model.language_model.layers.13.mlp.up_proj	58,982,400	4,097,766,656
model.language_model.layers.13.post_attention_layernorm	3,840	4,097,770,496
model.language_model.layers.13.post_feedforward_layernorm	3,840	4,097,774,336
model.language_model.layers.13.pre_feedforward_layernorm	3,840	4,097,778,176
model.language_model.layers.13.self_attn.k_norm	256	4,097,778,432
model.language_model.layers.13.self_attn.k_proj	7,864,320	4,105,642,752
model.language_model.layers.13.self_attn.o_proj	15,728,640	4,121,371,392
model.language_model.layers.13.self_attn.q_norm	256	4,121,371,648
model.language_model.layers.13.self_attn.q_proj	15,728,640	4,137,100,288
model.language_model.layers.13.self_attn.v_proj	7,864,320	4,144,964,608
model.language_model.layers.14.input_layernorm	3,840	4,144,968,448
model.language_model.layers.14.mlp.down_proj	58,982,400	4,203,950,848
model.language_model.layers.14.mlp.gate_proj	58,982,400	4,262,933,248
model.language_model.layers.14.mlp.up_proj	58,982,400	4,321,915,648
model.language_model.layers.14.post_attention_layernorm	3,840	4,321,919,488
model.language_model.layers.14.post_feedforward_layernorm	3,840	4,321,923,328
model.language_model.layers.14.pre_feedforward_layernorm	3,840	4,321,927,168
model.language_model.layers.14.self_attn.k_norm	256	4,321,927,424
model.language_model.layers.14.self_attn.k_proj	7,864,320	4,329,791,744
model.language_model.layers.14.self_attn.o_proj	15,728,640	4,345,520,384
model.language_model.layers.14.self_attn.q_norm	256	4,345,520,640
model.language_model.layers.14.self_attn.q_proj	15,728,640	4,361,249,280
model.language_model.layers.14.self_attn.v_proj	7,864,320	4,369,113,600
model.language_model.layers.15.input_layernorm	3,840	4,369,117,440
model.language_model.layers.15.mlp.down_proj	58,982,400	4,428,099,840
model.language_model.layers.15.mlp.gate_proj	58,982,400	4,487,082,240
model.language_model.layers.15.mlp.up_proj	58,982,400	4,546,064,640
model.language_model.layers.15.post_attention_layernorm	3,840	4,546,068,480
model.language_model.layers.15.post_feedforward_layernorm	3,840	4,546,072,320
model.language_model.layers.15.pre_feedforward_layernorm	3,840	4,546,076,160
model.language_model.layers.15.self_attn.k_norm	256	4,546,076,416
model.language_model.layers.15.self_attn.k_proj	7,864,320	4,553,940,736
model.language_model.layers.15.self_attn.o_proj	15,728,640	4,569,669,376
model.language_model.layers.15.self_attn.q_norm	256	4,569,669,632

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Module	Params	Running total
model.language_model.layers.15.self_attn.q_proj	15,728,640	4,585,398,272
model.language_model.layers.15.self_attn.v_proj	7,864,320	4,593,262,592
model.language_model.layers.16.input_layernorm	3,840	4,593,266,432
model.language_model.layers.16.mlp.down_proj	58,982,400	4,652,248,832
model.language_model.layers.16.mlp.gate_proj	58,982,400	4,711,231,232
model.language_model.layers.16.mlp.up_proj	58,982,400	4,770,213,632
model.language_model.layers.16.post_attention_layernorm	3,840	4,770,217,472
model.language_model.layers.16.post_feedforward_layernorm	3,840	4,770,221,312
model.language_model.layers.16.pre_feedforward_layernorm	3,840	4,770,225,152
model.language_model.layers.16.self_attn.k_norm	256	4,770,225,408
model.language_model.layers.16.self_attn.k_proj	7,864,320	4,778,089,728
model.language_model.layers.16.self_attn.o_proj	15,728,640	4,793,818,368
model.language_model.layers.16.self_attn.q_norm	256	4,793,818,624
model.language_model.layers.16.self_attn.q_proj	15,728,640	4,809,547,264
model.language_model.layers.16.self_attn.v_proj	7,864,320	4,817,411,584
model.language_model.layers.17.input_layernorm	3,840	4,817,415,424
model.language_model.layers.17.mlp.down_proj	58,982,400	4,876,397,824
model.language_model.layers.17.mlp.gate_proj	58,982,400	4,935,380,224
model.language_model.layers.17.mlp.up_proj	58,982,400	4,994,362,624
model.language_model.layers.17.post_attention_layernorm	3,840	4,994,366,464
model.language_model.layers.17.post_feedforward_layernorm	3,840	4,994,370,304
model.language_model.layers.17.pre_feedforward_layernorm	3,840	4,994,374,144
model.language_model.layers.17.self_attn.k_norm	256	4,994,374,400
model.language_model.layers.17.self_attn.k_proj	7,864,320	5,002,238,720
model.language_model.layers.17.self_attn.o_proj	15,728,640	5,017,967,360
model.language_model.layers.17.self_attn.q_norm	256	5,017,967,616
model.language_model.layers.17.self_attn.q_proj	15,728,640	5,033,696,256
model.language_model.layers.17.self_attn.v_proj	7,864,320	5,041,560,576
model.language_model.layers.18.input_layernorm	3,840	5,041,564,416
model.language_model.layers.18.mlp.down_proj	58,982,400	5,100,546,816
model.language_model.layers.18.mlp.gate_proj	58,982,400	5,159,529,216
model.language_model.layers.18.mlp.up_proj	58,982,400	5,218,511,616
model.language_model.layers.18.post_attention_layernorm	3,840	5,218,515,456
model.language_model.layers.18.post_feedforward_layernorm	3,840	5,218,519,296
model.language_model.layers.18.pre_feedforward_layernorm	3,840	5,218,523,136
model.language_model.layers.18.self_attn.k_norm	256	5,218,523,392
model.language_model.layers.18.self_attn.k_proj	7,864,320	5,226,387,712
model.language_model.layers.18.self_attn.o_proj	15,728,640	5,242,116,352
model.language_model.layers.18.self_attn.q_norm	256	5,242,116,608
model.language_model.layers.18.self_attn.q_proj	15,728,640	5,257,845,248
model.language_model.layers.18.self_attn.v_proj	7,864,320	5,265,709,568
model.language_model.layers.19.input_layernorm	3,840	5,265,713,408
model.language_model.layers.19.mlp.down_proj	58,982,400	5,324,695,808
model.language_model.layers.19.mlp.gate_proj	58,982,400	5,383,678,208
model.language_model.layers.19.mlp.up_proj	58,982,400	5,442,660,608
model.language_model.layers.19.post_attention_layernorm	3,840	5,442,664,448
model.language_model.layers.19.post_feedforward_layernorm	3,840	5,442,668,288

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Module	Params	Running total
model.language_model.layers.19. pre_feedforward_layernorm	3,840	5,442,672,128
model.language_model.layers.19.self_attn.k_norm	256	5,442,672,384
model.language_model.layers.19.self_attn.k_proj	7,864,320	5,450,536,704
model.language_model.layers.19.self_attn.o_proj	15,728,640	5,466,265,344
model.language_model.layers.19.self_attn.q_norm	256	5,466,265,600
model.language_model.layers.19.self_attn.q_proj	15,728,640	5,481,994,240
model.language_model.layers.19.self_attn.v_proj	7,864,320	5,489,858,560
model.language_model.layers.20.input_layernorm	3,840	5,489,862,400
model.language_model.layers.20.mlp.down_proj	58,982,400	5,548,844,800
model.language_model.layers.20.mlp.gate_proj	58,982,400	5,607,827,200
model.language_model.layers.20.mlp.up_proj	58,982,400	5,666,809,600
model.language_model.layers.20. post_attention_layernorm	3,840	5,666,813,440
model.language_model.layers.20. post_feedforward_layernorm	3,840	5,666,817,280
model.language_model.layers.20. pre_feedforward_layernorm	3,840	5,666,821,120
model.language_model.layers.20.self_attn.k_norm	256	5,666,821,376
model.language_model.layers.20.self_attn.k_proj	7,864,320	5,674,685,696
model.language_model.layers.20.self_attn.o_proj	15,728,640	5,690,414,336
model.language_model.layers.20.self_attn.q_norm	256	5,690,414,592
model.language_model.layers.20.self_attn.q_proj	15,728,640	5,706,143,232
model.language_model.layers.20.self_attn.v_proj	7,864,320	5,714,007,552
model.language_model.layers.21.input_layernorm	3,840	5,714,011,392
model.language_model.layers.21.mlp.down_proj	58,982,400	5,772,993,792
model.language_model.layers.21.mlp.gate_proj	58,982,400	5,831,976,192
model.language_model.layers.21.mlp.up_proj	58,982,400	5,890,958,592
model.language_model.layers.21. post_attention_layernorm	3,840	5,890,962,432
model.language_model.layers.21. post_feedforward_layernorm	3,840	5,890,966,272
model.language_model.layers.21. pre_feedforward_layernorm	3,840	5,890,970,112
model.language_model.layers.21.self_attn.k_norm	256	5,890,970,368
model.language_model.layers.21.self_attn.k_proj	7,864,320	5,898,834,688
model.language_model.layers.21.self_attn.o_proj	15,728,640	5,914,563,328
model.language_model.layers.21.self_attn.q_norm	256	5,914,563,584
model.language_model.layers.21.self_attn.q_proj	15,728,640	5,930,292,224
model.language_model.layers.21.self_attn.v_proj	7,864,320	5,938,156,544
model.language_model.layers.22.input_layernorm	3,840	5,938,160,384
model.language_model.layers.22.mlp.down_proj	58,982,400	5,997,142,784
model.language_model.layers.22.mlp.gate_proj	58,982,400	6,056,125,184
model.language_model.layers.22.mlp.up_proj	58,982,400	6,115,107,584
model.language_model.layers.22. post_attention_layernorm	3,840	6,115,111,424
model.language_model.layers.22. post_feedforward_layernorm	3,840	6,115,115,264
model.language_model.layers.22. pre_feedforward_layernorm	3,840	6,115,119,104
model.language_model.layers.22.self_attn.k_norm	256	6,115,119,360
model.language_model.layers.22.self_attn.k_proj	7,864,320	6,122,983,680
model.language_model.layers.22.self_attn.o_proj	15,728,640	6,138,712,320
model.language_model.layers.22.self_attn.q_norm	256	6,138,712,576
model.language_model.layers.22.self_attn.q_proj	15,728,640	6,154,441,216
model.language_model.layers.22.self_attn.v_proj	7,864,320	6,162,305,536
model.language_model.layers.23.input_layernorm	3,840	6,162,309,376
model.language_model.layers.23.mlp.down_proj	58,982,400	6,221,291,776

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Module	Params	Running total
model.language_model.layers.23.mlp.gate_proj	58,982,400	6,280,274,176
model.language_model.layers.23.mlp.up_proj	58,982,400	6,339,256,576
model.language_model.layers.23. post_attention_layernorm	3,840	6,339,260,416
model.language_model.layers.23. post_feedforward_layernorm	3,840	6,339,264,256
model.language_model.layers.23. pre_feedforward_layernorm	3,840	6,339,268,096
model.language_model.layers.23.self_attn.k_norm	256	6,339,268,352
model.language_model.layers.23.self_attn.k_proj	7,864,320	6,347,132,672
model.language_model.layers.23.self_attn.o_proj	15,728,640	6,362,861,312
model.language_model.layers.23.self_attn.q_norm	256	6,362,861,568
model.language_model.layers.23.self_attn.q_proj	15,728,640	6,378,590,208
model.language_model.layers.23.self_attn.v_proj	7,864,320	6,386,454,528
model.language_model.layers.24.input_layernorm	3,840	6,386,458,368
model.language_model.layers.24.mlp.down_proj	58,982,400	6,445,440,768
model.language_model.layers.24.mlp.gate_proj	58,982,400	6,504,423,168
model.language_model.layers.24.mlp.up_proj	58,982,400	6,563,405,568
model.language_model.layers.24. post_attention_layernorm	3,840	6,563,409,408
model.language_model.layers.24. post_feedforward_layernorm	3,840	6,563,413,248
model.language_model.layers.24. pre_feedforward_layernorm	3,840	6,563,417,088
model.language_model.layers.24.self_attn.k_norm	256	6,563,417,344
model.language_model.layers.24.self_attn.k_proj	7,864,320	6,571,281,664
model.language_model.layers.24.self_attn.o_proj	15,728,640	6,587,010,304
model.language_model.layers.24.self_attn.q_norm	256	6,587,010,560
model.language_model.layers.24.self_attn.q_proj	15,728,640	6,602,739,200
model.language_model.layers.24.self_attn.v_proj	7,864,320	6,610,603,520
model.language_model.layers.25.input_layernorm	3,840	6,610,607,360
model.language_model.layers.25.mlp.down_proj	58,982,400	6,669,589,760
model.language_model.layers.25.mlp.gate_proj	58,982,400	6,728,572,160
model.language_model.layers.25.mlp.up_proj	58,982,400	6,787,554,560
model.language_model.layers.25. post_attention_layernorm	3,840	6,787,558,400
model.language_model.layers.25. post_feedforward_layernorm	3,840	6,787,562,240
model.language_model.layers.25. pre_feedforward_layernorm	3,840	6,787,566,080
model.language_model.layers.25.self_attn.k_norm	256	6,787,566,336
model.language_model.layers.25.self_attn.k_proj	7,864,320	6,795,430,656
model.language_model.layers.25.self_attn.o_proj	15,728,640	6,811,159,296
model.language_model.layers.25.self_attn.q_norm	256	6,811,159,552
model.language_model.layers.25.self_attn.q_proj	15,728,640	6,826,888,192
model.language_model.layers.25.self_attn.v_proj	7,864,320	6,834,752,512
model.language_model.layers.26.input_layernorm	3,840	6,834,756,352
model.language_model.layers.26.mlp.down_proj	58,982,400	6,893,738,752
model.language_model.layers.26.mlp.gate_proj	58,982,400	6,952,721,152
model.language_model.layers.26.mlp.up_proj	58,982,400	7,011,703,552
model.language_model.layers.26. post_attention_layernorm	3,840	7,011,707,392
model.language_model.layers.26. post_feedforward_layernorm	3,840	7,011,711,232
model.language_model.layers.26. pre_feedforward_layernorm	3,840	7,011,715,072
model.language_model.layers.26.self_attn.k_norm	256	7,011,715,328
model.language_model.layers.26.self_attn.k_proj	7,864,320	7,019,579,648

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Module	Params	Running total
model.language_model.layers.26.self_attn.o_proj	15,728,640	7,035,308,288
model.language_model.layers.26.self_attn.q_norm	256	7,035,308,544
model.language_model.layers.26.self_attn.q_proj	15,728,640	7,051,037,184
model.language_model.layers.26.self_attn.v_proj	7,864,320	7,058,901,504
model.language_model.layers.27.input_layernorm	3,840	7,058,905,344
model.language_model.layers.27.mlp.down_proj	58,982,400	7,117,887,744
model.language_model.layers.27.mlp.gate_proj	58,982,400	7,176,870,144
model.language_model.layers.27.mlp.up_proj	58,982,400	7,235,852,544
model.language_model.layers.27.post_attention_layernorm	3,840	7,235,856,384
model.language_model.layers.27.post_feedforward_layernorm	3,840	7,235,860,224
model.language_model.layers.27.pre_feedforward_layernorm	3,840	7,235,864,064
model.language_model.layers.27.self_attn.k_norm	256	7,235,864,320
model.language_model.layers.27.self_attn.k_proj	7,864,320	7,243,728,640
model.language_model.layers.27.self_attn.o_proj	15,728,640	7,259,457,280
model.language_model.layers.27.self_attn.q_norm	256	7,259,457,536
model.language_model.layers.27.self_attn.q_proj	15,728,640	7,275,186,176
model.language_model.layers.27.self_attn.v_proj	7,864,320	7,283,050,496
model.language_model.layers.28.input_layernorm	3,840	7,283,054,336
model.language_model.layers.28.mlp.down_proj	58,982,400	7,342,036,736
model.language_model.layers.28.mlp.gate_proj	58,982,400	7,401,019,136
model.language_model.layers.28.mlp.up_proj	58,982,400	7,460,001,536
model.language_model.layers.28.post_attention_layernorm	3,840	7,460,005,376
model.language_model.layers.28.post_feedforward_layernorm	3,840	7,460,009,216
model.language_model.layers.28.pre_feedforward_layernorm	3,840	7,460,013,056
model.language_model.layers.28.self_attn.k_norm	256	7,460,013,312
model.language_model.layers.28.self_attn.k_proj	7,864,320	7,467,877,632
model.language_model.layers.28.self_attn.o_proj	15,728,640	7,483,606,272
model.language_model.layers.28.self_attn.q_norm	256	7,483,606,528
model.language_model.layers.28.self_attn.q_proj	15,728,640	7,499,335,168
model.language_model.layers.28.self_attn.v_proj	7,864,320	7,507,199,488
model.language_model.layers.29.input_layernorm	3,840	7,507,203,328
model.language_model.layers.29.mlp.down_proj	58,982,400	7,566,185,728
model.language_model.layers.29.mlp.gate_proj	58,982,400	7,625,168,128
model.language_model.layers.29.mlp.up_proj	58,982,400	7,684,150,528
model.language_model.layers.29.post_attention_layernorm	3,840	7,684,154,368
model.language_model.layers.29.post_feedforward_layernorm	3,840	7,684,158,208
model.language_model.layers.29.pre_feedforward_layernorm	3,840	7,684,162,048
model.language_model.layers.29.self_attn.k_norm	256	7,684,162,304
model.language_model.layers.29.self_attn.k_proj	7,864,320	7,692,026,624
model.language_model.layers.29.self_attn.o_proj	15,728,640	7,707,755,264
model.language_model.layers.29.self_attn.q_norm	256	7,707,755,520
model.language_model.layers.29.self_attn.q_proj	15,728,640	7,723,484,160
model.language_model.layers.29.self_attn.v_proj	7,864,320	7,731,348,480
model.language_model.layers.30.input_layernorm	3,840	7,731,352,320
model.language_model.layers.30.mlp.down_proj	58,982,400	7,790,334,720
model.language_model.layers.30.mlp.gate_proj	58,982,400	7,849,317,120
model.language_model.layers.30.mlp.up_proj	58,982,400	7,908,299,520
model.language_model.layers.30.post_attention_layernorm	3,840	7,908,303,360

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Module	Params	Running total
model.language_model.layers.30. post_feedforward_layernorm	3,840	7,908,307,200
model.language_model.layers.30. pre_feedforward_layernorm	3,840	7,908,311,040
model.language_model.layers.30.self_attn.k_norm	256	7,908,311,296
model.language_model.layers.30.self_attn.k_proj	7,864,320	7,916,175,616
model.language_model.layers.30.self_attn.o_proj	15,728,640	7,931,904,256
model.language_model.layers.30.self_attn.q_norm	256	7,931,904,512
model.language_model.layers.30.self_attn.q_proj	15,728,640	7,947,633,152
model.language_model.layers.30.self_attn.v_proj	7,864,320	7,955,497,472
model.language_model.layers.31.input_layernorm	3,840	7,955,501,312
model.language_model.layers.31.mlp.down_proj	58,982,400	8,014,483,712
model.language_model.layers.31.mlp.gate_proj	58,982,400	8,073,466,112
model.language_model.layers.31.mlp.up_proj	58,982,400	8,132,448,512
model.language_model.layers.31. post_attention_layernorm	3,840	8,132,452,352
model.language_model.layers.31. post_feedforward_layernorm	3,840	8,132,456,192
model.language_model.layers.31. pre_feedforward_layernorm	3,840	8,132,460,032
model.language_model.layers.31.self_attn.k_norm	256	8,132,460,288
model.language_model.layers.31.self_attn.k_proj	7,864,320	8,140,324,608
model.language_model.layers.31.self_attn.o_proj	15,728,640	8,156,053,248
model.language_model.layers.31.self_attn.q_norm	256	8,156,053,504
model.language_model.layers.31.self_attn.q_proj	15,728,640	8,171,782,144
model.language_model.layers.31.self_attn.v_proj	7,864,320	8,179,646,464
model.language_model.layers.32.input_layernorm	3,840	8,179,650,304
model.language_model.layers.32.mlp.down_proj	58,982,400	8,238,632,704
model.language_model.layers.32.mlp.gate_proj	58,982,400	8,297,615,104
model.language_model.layers.32.mlp.up_proj	58,982,400	8,356,597,504
model.language_model.layers.32. post_attention_layernorm	3,840	8,356,601,344
model.language_model.layers.32. post_feedforward_layernorm	3,840	8,356,605,184
model.language_model.layers.32. pre_feedforward_layernorm	3,840	8,356,609,024
model.language_model.layers.32.self_attn.k_norm	256	8,356,609,280
model.language_model.layers.32.self_attn.k_proj	7,864,320	8,364,473,600
model.language_model.layers.32.self_attn.o_proj	15,728,640	8,380,202,240
model.language_model.layers.32.self_attn.q_norm	256	8,380,202,496
model.language_model.layers.32.self_attn.q_proj	15,728,640	8,395,931,136
model.language_model.layers.32.self_attn.v_proj	7,864,320	8,403,795,456
model.language_model.layers.33.input_layernorm	3,840	8,403,799,296
model.language_model.layers.33.mlp.down_proj	58,982,400	8,462,781,696
model.language_model.layers.33.mlp.gate_proj	58,982,400	8,521,764,096
model.language_model.layers.33.mlp.up_proj	58,982,400	8,580,746,496
model.language_model.layers.33. post_attention_layernorm	3,840	8,580,750,336
model.language_model.layers.33. post_feedforward_layernorm	3,840	8,580,754,176
model.language_model.layers.33. pre_feedforward_layernorm	3,840	8,580,758,016
model.language_model.layers.33.self_attn.k_norm	256	8,580,758,272
model.language_model.layers.33.self_attn.k_proj	7,864,320	8,588,622,592
model.language_model.layers.33.self_attn.o_proj	15,728,640	8,604,351,232
model.language_model.layers.33.self_attn.q_norm	256	8,604,351,488
model.language_model.layers.33.self_attn.q_proj	15,728,640	8,620,080,128
model.language_model.layers.33.self_attn.v_proj	7,864,320	8,627,944,448

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Module	Params	Running total
model.language_model.layers.34.input_layernorm	3,840	8,627,948,288
model.language_model.layers.34.mlp.down_proj	58,982,400	8,686,930,688
model.language_model.layers.34.mlp.gate_proj	58,982,400	8,745,913,088
model.language_model.layers.34.mlp.up_proj	58,982,400	8,804,895,488
model.language_model.layers.34.post_attention_layernorm	3,840	8,804,899,328
model.language_model.layers.34.post_feedforward_layernorm	3,840	8,804,903,168
model.language_model.layers.34.pre_feedforward_layernorm	3,840	8,804,907,008
model.language_model.layers.34.self_attn.k_norm	256	8,804,907,264
model.language_model.layers.34.self_attn.k_proj	7,864,320	8,812,771,584
model.language_model.layers.34.self_attn.o_proj	15,728,640	8,828,500,224
model.language_model.layers.34.self_attn.q_norm	256	8,828,500,480
model.language_model.layers.34.self_attn.q_proj	15,728,640	8,844,229,120
model.language_model.layers.34.self_attn.v_proj	7,864,320	8,852,093,440
model.language_model.layers.35.input_layernorm	3,840	8,852,097,280
model.language_model.layers.35.mlp.down_proj	58,982,400	8,911,079,680
model.language_model.layers.35.mlp.gate_proj	58,982,400	8,970,062,080
model.language_model.layers.35.mlp.up_proj	58,982,400	9,029,044,480
model.language_model.layers.35.post_attention_layernorm	3,840	9,029,048,320
model.language_model.layers.35.post_feedforward_layernorm	3,840	9,029,052,160
model.language_model.layers.35.pre_feedforward_layernorm	3,840	9,029,056,000
model.language_model.layers.35.self_attn.k_norm	256	9,029,056,256
model.language_model.layers.35.self_attn.k_proj	7,864,320	9,036,920,576
model.language_model.layers.35.self_attn.o_proj	15,728,640	9,052,649,216
model.language_model.layers.35.self_attn.q_norm	256	9,052,649,472
model.language_model.layers.35.self_attn.q_proj	15,728,640	9,068,378,112
model.language_model.layers.35.self_attn.v_proj	7,864,320	9,076,242,432
model.language_model.layers.36.input_layernorm	3,840	9,076,246,272
model.language_model.layers.36.mlp.down_proj	58,982,400	9,135,228,672
model.language_model.layers.36.mlp.gate_proj	58,982,400	9,194,211,072
model.language_model.layers.36.mlp.up_proj	58,982,400	9,253,193,472
model.language_model.layers.36.post_attention_layernorm	3,840	9,253,197,312
model.language_model.layers.36.post_feedforward_layernorm	3,840	9,253,201,152
model.language_model.layers.36.pre_feedforward_layernorm	3,840	9,253,204,992
model.language_model.layers.36.self_attn.k_norm	256	9,253,205,248
model.language_model.layers.36.self_attn.k_proj	7,864,320	9,261,069,568
model.language_model.layers.36.self_attn.o_proj	15,728,640	9,276,798,208
model.language_model.layers.36.self_attn.q_norm	256	9,276,798,464
model.language_model.layers.36.self_attn.q_proj	15,728,640	9,292,527,104
model.language_model.layers.36.self_attn.v_proj	7,864,320	9,300,391,424
model.language_model.layers.37.input_layernorm	3,840	9,300,395,264
model.language_model.layers.37.mlp.down_proj	58,982,400	9,359,377,664
model.language_model.layers.37.mlp.gate_proj	58,982,400	9,418,360,064
model.language_model.layers.37.mlp.up_proj	58,982,400	9,477,342,464
model.language_model.layers.37.post_attention_layernorm	3,840	9,477,346,304
model.language_model.layers.37.post_feedforward_layernorm	3,840	9,477,350,144
model.language_model.layers.37.pre_feedforward_layernorm	3,840	9,477,353,984

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Module	Params	Running total
model.language_model.layers.37.self_attn.k_norm	256	9,477,354,240
model.language_model.layers.37.self_attn.k_proj	7,864,320	9,485,218,560
model.language_model.layers.37.self_attn.o_proj	15,728,640	9,500,947,200
model.language_model.layers.37.self_attn.q_norm	256	9,500,947,456
model.language_model.layers.37.self_attn.q_proj	15,728,640	9,516,676,096
model.language_model.layers.37.self_attn.v_proj	7,864,320	9,524,540,416
model.language_model.layers.38.input_layernorm	3,840	9,524,544,256
model.language_model.layers.38.mlp.down_proj	58,982,400	9,583,526,656
model.language_model.layers.38.mlp.gate_proj	58,982,400	9,642,509,056
model.language_model.layers.38.mlp.up_proj	58,982,400	9,701,491,456
model.language_model.layers.38.post_attention_layernorm	3,840	9,701,495,296
model.language_model.layers.38.post_feedforward_layernorm	3,840	9,701,499,136
model.language_model.layers.38.pre_feedforward_layernorm	3,840	9,701,502,976
model.language_model.layers.38.self_attn.k_norm	256	9,701,503,232
model.language_model.layers.38.self_attn.k_proj	7,864,320	9,709,367,552
model.language_model.layers.38.self_attn.o_proj	15,728,640	9,725,096,192
model.language_model.layers.38.self_attn.q_norm	256	9,725,096,448
model.language_model.layers.38.self_attn.q_proj	15,728,640	9,740,825,088
model.language_model.layers.38.self_attn.v_proj	7,864,320	9,748,689,408
model.language_model.layers.39.input_layernorm	3,840	9,748,693,248
model.language_model.layers.39.mlp.down_proj	58,982,400	9,807,675,648
model.language_model.layers.39.mlp.gate_proj	58,982,400	9,866,658,048
model.language_model.layers.39.mlp.up_proj	58,982,400	9,925,640,448
model.language_model.layers.39.post_attention_layernorm	3,840	9,925,644,288
model.language_model.layers.39.post_feedforward_layernorm	3,840	9,925,648,128
model.language_model.layers.39.pre_feedforward_layernorm	3,840	9,925,651,968
model.language_model.layers.39.self_attn.k_norm	256	9,925,652,224
model.language_model.layers.39.self_attn.k_proj	7,864,320	9,933,516,544
model.language_model.layers.39.self_attn.o_proj	15,728,640	9,949,245,184
model.language_model.layers.39.self_attn.q_norm	256	9,949,245,440
model.language_model.layers.39.self_attn.q_proj	15,728,640	9,964,974,080
model.language_model.layers.39.self_attn.v_proj	7,864,320	9,972,838,400
model.language_model.layers.40.input_layernorm	3,840	9,972,842,240
model.language_model.layers.40.mlp.down_proj	58,982,400	10,031,824,640
model.language_model.layers.40.mlp.gate_proj	58,982,400	10,090,807,040
model.language_model.layers.40.mlp.up_proj	58,982,400	10,149,789,440
model.language_model.layers.40.post_attention_layernorm	3,840	10,149,793,280
model.language_model.layers.40.post_feedforward_layernorm	3,840	10,149,797,120
model.language_model.layers.40.pre_feedforward_layernorm	3,840	10,149,800,960
model.language_model.layers.40.self_attn.k_norm	256	10,149,801,216
model.language_model.layers.40.self_attn.k_proj	7,864,320	10,157,665,536
model.language_model.layers.40.self_attn.o_proj	15,728,640	10,173,394,176
model.language_model.layers.40.self_attn.q_norm	256	10,173,394,432
model.language_model.layers.40.self_attn.q_proj	15,728,640	10,189,123,072
model.language_model.layers.40.self_attn.v_proj	7,864,320	10,196,987,392
model.language_model.layers.41.input_layernorm	3,840	10,196,991,232
model.language_model.layers.41.mlp.down_proj	58,982,400	10,255,973,632
model.language_model.layers.41.mlp.gate_proj	58,982,400	10,314,956,032
model.language_model.layers.41.mlp.up_proj	58,982,400	10,373,938,432

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Module	Params	Running total
model.language_model.layers.41. post_attention_layernorm	3,840	10,373,942,272
model.language_model.layers.41. post_feedforward_layernorm	3,840	10,373,946,112
model.language_model.layers.41. pre_feedforward_layernorm	3,840	10,373,949,952
model.language_model.layers.41.self_attn.k_norm	256	10,373,950,208
model.language_model.layers.41.self_attn.k_proj	7,864,320	10,381,814,528
model.language_model.layers.41.self_attn.o_proj	15,728,640	10,397,543,168
model.language_model.layers.41.self_attn.q_norm	256	10,397,543,424
model.language_model.layers.41.self_attn.q_proj	15,728,640	10,413,272,064
model.language_model.layers.41.self_attn.v_proj	7,864,320	10,421,136,384
model.language_model.layers.42.input_layernorm	3,840	10,421,140,224
model.language_model.layers.42.mlp.down_proj	58,982,400	10,480,122,624
model.language_model.layers.42.mlp.gate_proj	58,982,400	10,539,105,024
model.language_model.layers.42.mlp.up_proj	58,982,400	10,598,087,424
model.language_model.layers.42. post_attention_layernorm	3,840	10,598,091,264
model.language_model.layers.42. post_feedforward_layernorm	3,840	10,598,095,104
model.language_model.layers.42. pre_feedforward_layernorm	3,840	10,598,098,944
model.language_model.layers.42.self_attn.k_norm	256	10,598,099,200
model.language_model.layers.42.self_attn.k_proj	7,864,320	10,605,963,520
model.language_model.layers.42.self_attn.o_proj	15,728,640	10,621,692,160
model.language_model.layers.42.self_attn.q_norm	256	10,621,692,416
model.language_model.layers.42.self_attn.q_proj	15,728,640	10,637,421,056
model.language_model.layers.42.self_attn.v_proj	7,864,320	10,645,285,376
model.language_model.layers.43.input_layernorm	3,840	10,645,289,216
model.language_model.layers.43.mlp.down_proj	58,982,400	10,704,271,616
model.language_model.layers.43.mlp.gate_proj	58,982,400	10,763,254,016
model.language_model.layers.43.mlp.up_proj	58,982,400	10,822,236,416
model.language_model.layers.43. post_attention_layernorm	3,840	10,822,240,256
model.language_model.layers.43. post_feedforward_layernorm	3,840	10,822,244,096
model.language_model.layers.43. pre_feedforward_layernorm	3,840	10,822,247,936
model.language_model.layers.43.self_attn.k_norm	256	10,822,248,192
model.language_model.layers.43.self_attn.k_proj	7,864,320	10,830,112,512
model.language_model.layers.43.self_attn.o_proj	15,728,640	10,845,841,152
model.language_model.layers.43.self_attn.q_norm	256	10,845,841,408
model.language_model.layers.43.self_attn.q_proj	15,728,640	10,861,570,048
model.language_model.layers.43.self_attn.v_proj	7,864,320	10,869,434,368
model.language_model.layers.44.input_layernorm	3,840	10,869,438,208
model.language_model.layers.44.mlp.down_proj	58,982,400	10,928,420,608
model.language_model.layers.44.mlp.gate_proj	58,982,400	10,987,403,008
model.language_model.layers.44.mlp.up_proj	58,982,400	11,046,385,408
model.language_model.layers.44. post_attention_layernorm	3,840	11,046,389,248
model.language_model.layers.44. post_feedforward_layernorm	3,840	11,046,393,088
model.language_model.layers.44. pre_feedforward_layernorm	3,840	11,046,396,928
model.language_model.layers.44.self_attn.k_norm	256	11,046,397,184
model.language_model.layers.44.self_attn.k_proj	7,864,320	11,054,261,504
model.language_model.layers.44.self_attn.o_proj	15,728,640	11,069,990,144
model.language_model.layers.44.self_attn.q_norm	256	11,069,990,400

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Module	Params	Running total
model.language_model.layers.44.self_attn.q_proj	15,728,640	11,085,719,040
model.language_model.layers.44.self_attn.v_proj	7,864,320	11,093,583,360
model.language_model.layers.45.input_layernorm	3,840	11,093,587,200
model.language_model.layers.45.mlp.down_proj	58,982,400	11,152,569,600
model.language_model.layers.45.mlp.gate_proj	58,982,400	11,211,552,000
model.language_model.layers.45.mlp.up_proj	58,982,400	11,270,534,400
model.language_model.layers.45.post_attention_layernorm	3,840	11,270,538,240
model.language_model.layers.45.post_feedforward_layernorm	3,840	11,270,542,080
model.language_model.layers.45.pre_feedforward_layernorm	3,840	11,270,545,920
model.language_model.layers.45.self_attn.k_norm	256	11,270,546,176
model.language_model.layers.45.self_attn.k_proj	7,864,320	11,278,410,496
model.language_model.layers.45.self_attn.o_proj	15,728,640	11,294,139,136
model.language_model.layers.45.self_attn.q_norm	256	11,294,139,392
model.language_model.layers.45.self_attn.q_proj	15,728,640	11,309,868,032
model.language_model.layers.45.self_attn.v_proj	7,864,320	11,317,732,352
model.language_model.layers.46.input_layernorm	3,840	11,317,736,192
model.language_model.layers.46.mlp.down_proj	58,982,400	11,376,718,592
model.language_model.layers.46.mlp.gate_proj	58,982,400	11,435,700,992
model.language_model.layers.46.mlp.up_proj	58,982,400	11,494,683,392
model.language_model.layers.46.post_attention_layernorm	3,840	11,494,687,232
model.language_model.layers.46.post_feedforward_layernorm	3,840	11,494,691,072
model.language_model.layers.46.pre_feedforward_layernorm	3,840	11,494,694,912
model.language_model.layers.46.self_attn.k_norm	256	11,494,695,168
model.language_model.layers.46.self_attn.k_proj	7,864,320	11,502,559,488
model.language_model.layers.46.self_attn.o_proj	15,728,640	11,518,288,128
model.language_model.layers.46.self_attn.q_norm	256	11,518,288,384
model.language_model.layers.46.self_attn.q_proj	15,728,640	11,534,017,024
model.language_model.layers.46.self_attn.v_proj	7,864,320	11,541,881,344
model.language_model.layers.47.input_layernorm	3,840	11,541,885,184
model.language_model.layers.47.mlp.down_proj	58,982,400	11,600,867,584
model.language_model.layers.47.mlp.gate_proj	58,982,400	11,659,849,984
model.language_model.layers.47.mlp.up_proj	58,982,400	11,718,832,384
model.language_model.layers.47.post_attention_layernorm	3,840	11,718,836,224
model.language_model.layers.47.post_feedforward_layernorm	3,840	11,718,840,064
model.language_model.layers.47.pre_feedforward_layernorm	3,840	11,718,843,904
model.language_model.layers.47.self_attn.k_norm	256	11,718,844,160
model.language_model.layers.47.self_attn.k_proj	7,864,320	11,726,708,480
model.language_model.layers.47.self_attn.o_proj	15,728,640	11,742,437,120
model.language_model.layers.47.self_attn.q_norm	256	11,742,437,376
model.language_model.layers.47.self_attn.q_proj	15,728,640	11,758,166,016
model.language_model.layers.47.self_attn.v_proj	7,864,320	11,766,030,336
model.language_model.norm	3,840	11,766,034,176
model.multi_modal_projector	4,423,680	11,770,457,856
model.multi_modal_projector.mm_soft_emb_norm	1,152	11,770,459,008
model.vision_tower.embeddings.patch_embedding	678,528	11,771,137,536
model.vision_tower.embeddings.position_embedding	4,718,592	11,775,856,128
model.vision_tower.encoder.layers.0.layer_norm1	2,304	11,775,858,432
model.vision_tower.encoder.layers.0.layer_norm2	2,304	11,775,860,736
model.vision_tower.encoder.layers.0.mlp.fc1	4,962,512	11,780,823,248

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Module	Params	Running total
model.vision_tower.encoder.layers.0.mlp.fc2	4,959,360	11,785,782,608
model.vision_tower.encoder.layers.0.self_attn.k_proj	1,328,256	11,787,110,864
model.vision_tower.encoder.layers.0.self_attn.out_proj	1,328,256	11,788,439,120
model.vision_tower.encoder.layers.0.self_attn.q_proj	1,328,256	11,789,767,376
model.vision_tower.encoder.layers.0.self_attn.v_proj	1,328,256	11,791,095,632
model.vision_tower.encoder.layers.1.layer_norm1	2,304	11,791,097,936
model.vision_tower.encoder.layers.1.layer_norm2	2,304	11,791,100,240
model.vision_tower.encoder.layers.1.mlp.fc1	4,962,512	11,796,062,752
model.vision_tower.encoder.layers.1.mlp.fc2	4,959,360	11,801,022,112
model.vision_tower.encoder.layers.1.self_attn.k_proj	1,328,256	11,802,350,368
model.vision_tower.encoder.layers.1.self_attn.out_proj	1,328,256	11,803,678,624
model.vision_tower.encoder.layers.1.self_attn.q_proj	1,328,256	11,805,006,880
model.vision_tower.encoder.layers.1.self_attn.v_proj	1,328,256	11,806,335,136
model.vision_tower.encoder.layers.2.layer_norm1	2,304	11,806,337,440
model.vision_tower.encoder.layers.2.layer_norm2	2,304	11,806,339,744
model.vision_tower.encoder.layers.2.mlp.fc1	4,962,512	11,811,302,256
model.vision_tower.encoder.layers.2.mlp.fc2	4,959,360	11,816,261,616
model.vision_tower.encoder.layers.2.self_attn.k_proj	1,328,256	11,817,589,872
model.vision_tower.encoder.layers.2.self_attn.out_proj	1,328,256	11,818,918,128
model.vision_tower.encoder.layers.2.self_attn.q_proj	1,328,256	11,820,246,384
model.vision_tower.encoder.layers.2.self_attn.v_proj	1,328,256	11,821,574,640
model.vision_tower.encoder.layers.3.layer_norm1	2,304	11,821,576,944
model.vision_tower.encoder.layers.3.layer_norm2	2,304	11,821,579,248
model.vision_tower.encoder.layers.3.mlp.fc1	4,962,512	11,826,541,760
model.vision_tower.encoder.layers.3.mlp.fc2	4,959,360	11,831,501,120
model.vision_tower.encoder.layers.3.self_attn.k_proj	1,328,256	11,832,829,376
model.vision_tower.encoder.layers.3.self_attn.out_proj	1,328,256	11,834,157,632
model.vision_tower.encoder.layers.3.self_attn.q_proj	1,328,256	11,835,485,888
model.vision_tower.encoder.layers.3.self_attn.v_proj	1,328,256	11,836,814,144
model.vision_tower.encoder.layers.4.layer_norm1	2,304	11,836,816,448
model.vision_tower.encoder.layers.4.layer_norm2	2,304	11,836,818,752
model.vision_tower.encoder.layers.4.mlp.fc1	4,962,512	11,841,781,264
model.vision_tower.encoder.layers.4.mlp.fc2	4,959,360	11,846,740,624
model.vision_tower.encoder.layers.4.self_attn.k_proj	1,328,256	11,848,068,880
model.vision_tower.encoder.layers.4.self_attn.out_proj	1,328,256	11,849,397,136
model.vision_tower.encoder.layers.4.self_attn.q_proj	1,328,256	11,850,725,392
model.vision_tower.encoder.layers.4.self_attn.v_proj	1,328,256	11,852,053,648
model.vision_tower.encoder.layers.5.layer_norm1	2,304	11,852,055,952

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Module	Params	Running total
model.vision_tower.encoder.layers.5.layer_norm2	2,304	11,852,058,256
model.vision_tower.encoder.layers.5.mlp.fc1	4,962,512	11,857,020,768
model.vision_tower.encoder.layers.5.mlp.fc2	4,959,360	11,861,980,128
model.vision_tower.encoder.layers.5.self_attn.k_proj	1,328,256	11,863,308,384
model.vision_tower.encoder.layers.5.self_attn.out_proj	1,328,256	11,864,636,640
model.vision_tower.encoder.layers.5.self_attn.q_proj	1,328,256	11,865,964,896
model.vision_tower.encoder.layers.5.self_attn.v_proj	1,328,256	11,867,293,152
model.vision_tower.encoder.layers.6.layer_norm1	2,304	11,867,295,456
model.vision_tower.encoder.layers.6.layer_norm2	2,304	11,867,297,760
model.vision_tower.encoder.layers.6.mlp.fc1	4,962,512	11,872,260,272
model.vision_tower.encoder.layers.6.mlp.fc2	4,959,360	11,877,219,632
model.vision_tower.encoder.layers.6.self_attn.k_proj	1,328,256	11,878,547,888
model.vision_tower.encoder.layers.6.self_attn.out_proj	1,328,256	11,879,876,144
model.vision_tower.encoder.layers.6.self_attn.q_proj	1,328,256	11,881,204,400
model.vision_tower.encoder.layers.6.self_attn.v_proj	1,328,256	11,882,532,656
model.vision_tower.encoder.layers.7.layer_norm1	2,304	11,882,534,960
model.vision_tower.encoder.layers.7.layer_norm2	2,304	11,882,537,264
model.vision_tower.encoder.layers.7.mlp.fc1	4,962,512	11,887,499,776
model.vision_tower.encoder.layers.7.mlp.fc2	4,959,360	11,892,459,136
model.vision_tower.encoder.layers.7.self_attn.k_proj	1,328,256	11,893,787,392
model.vision_tower.encoder.layers.7.self_attn.out_proj	1,328,256	11,895,115,648
model.vision_tower.encoder.layers.7.self_attn.q_proj	1,328,256	11,896,443,904
model.vision_tower.encoder.layers.7.self_attn.v_proj	1,328,256	11,897,772,160
model.vision_tower.encoder.layers.8.layer_norm1	2,304	11,897,774,464
model.vision_tower.encoder.layers.8.layer_norm2	2,304	11,897,776,768
model.vision_tower.encoder.layers.8.mlp.fc1	4,962,512	11,902,739,280
model.vision_tower.encoder.layers.8.mlp.fc2	4,959,360	11,907,698,640
model.vision_tower.encoder.layers.8.self_attn.k_proj	1,328,256	11,909,026,896
model.vision_tower.encoder.layers.8.self_attn.out_proj	1,328,256	11,910,355,152
model.vision_tower.encoder.layers.8.self_attn.q_proj	1,328,256	11,911,683,408
model.vision_tower.encoder.layers.8.self_attn.v_proj	1,328,256	11,913,011,664
model.vision_tower.encoder.layers.9.layer_norm1	2,304	11,913,013,968
model.vision_tower.encoder.layers.9.layer_norm2	2,304	11,913,016,272
model.vision_tower.encoder.layers.9.mlp.fc1	4,962,512	11,917,978,784
model.vision_tower.encoder.layers.9.mlp.fc2	4,959,360	11,922,938,144
model.vision_tower.encoder.layers.9.self_attn.k_proj	1,328,256	11,924,266,400
model.vision_tower.encoder.layers.9.self_attn.out_proj	1,328,256	11,925,594,656
model.vision_tower.encoder.layers.9.self_attn.q_proj	1,328,256	11,926,922,912

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Module	Params	Running total
model.vision_tower.encoder.layers.9.self_attn.v_proj	1,328,256	11,928,251,168
model.vision_tower.encoder.layers.10.layer_norm1	2,304	11,928,253,472
model.vision_tower.encoder.layers.10.layer_norm2	2,304	11,928,255,776
model.vision_tower.encoder.layers.10.mlp.fc1	4,962,512	11,933,218,288
model.vision_tower.encoder.layers.10.mlp.fc2	4,959,360	11,938,177,648
model.vision_tower.encoder.layers.10.self_attn.k_proj	1,328,256	11,939,505,904
model.vision_tower.encoder.layers.10.self_attn.out_proj	1,328,256	11,940,834,160
model.vision_tower.encoder.layers.10.self_attn.q_proj	1,328,256	11,942,162,416
model.vision_tower.encoder.layers.10.self_attn.v_proj	1,328,256	11,943,490,672
model.vision_tower.encoder.layers.11.layer_norm1	2,304	11,943,492,976
model.vision_tower.encoder.layers.11.layer_norm2	2,304	11,943,495,280
model.vision_tower.encoder.layers.11.mlp.fc1	4,962,512	11,948,457,792
model.vision_tower.encoder.layers.11.mlp.fc2	4,959,360	11,953,417,152
model.vision_tower.encoder.layers.11.self_attn.k_proj	1,328,256	11,954,745,408
model.vision_tower.encoder.layers.11.self_attn.out_proj	1,328,256	11,956,073,664
model.vision_tower.encoder.layers.11.self_attn.q_proj	1,328,256	11,957,401,920
model.vision_tower.encoder.layers.11.self_attn.v_proj	1,328,256	11,958,730,176
model.vision_tower.encoder.layers.12.layer_norm1	2,304	11,958,732,480
model.vision_tower.encoder.layers.12.layer_norm2	2,304	11,958,734,784
model.vision_tower.encoder.layers.12.mlp.fc1	4,962,512	11,963,697,296
model.vision_tower.encoder.layers.12.mlp.fc2	4,959,360	11,968,656,656
model.vision_tower.encoder.layers.12.self_attn.k_proj	1,328,256	11,969,984,912
model.vision_tower.encoder.layers.12.self_attn.out_proj	1,328,256	11,971,313,168
model.vision_tower.encoder.layers.12.self_attn.q_proj	1,328,256	11,972,641,424
model.vision_tower.encoder.layers.12.self_attn.v_proj	1,328,256	11,973,969,680
model.vision_tower.encoder.layers.13.layer_norm1	2,304	11,973,971,984
model.vision_tower.encoder.layers.13.layer_norm2	2,304	11,973,974,288
model.vision_tower.encoder.layers.13.mlp.fc1	4,962,512	11,978,936,800
model.vision_tower.encoder.layers.13.mlp.fc2	4,959,360	11,983,896,160
model.vision_tower.encoder.layers.13.self_attn.k_proj	1,328,256	11,985,224,416
model.vision_tower.encoder.layers.13.self_attn.out_proj	1,328,256	11,986,552,672
model.vision_tower.encoder.layers.13.self_attn.q_proj	1,328,256	11,987,880,928
model.vision_tower.encoder.layers.13.self_attn.v_proj	1,328,256	11,989,209,184
model.vision_tower.encoder.layers.14.layer_norm1	2,304	11,989,211,488
model.vision_tower.encoder.layers.14.layer_norm2	2,304	11,989,213,792
model.vision_tower.encoder.layers.14.mlp.fc1	4,962,512	11,994,176,304
model.vision_tower.encoder.layers.14.mlp.fc2	4,959,360	11,999,135,664
model.vision_tower.encoder.layers.14.self_attn.k_proj	1,328,256	12,000,463,920
model.vision_tower.encoder.layers.14.self_attn.out_proj	1,328,256	12,001,792,176

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Module	Params	Running total
model.vision_tower.encoder.layers.14.self_attn.q_proj	1,328,256	12,003,120,432
model.vision_tower.encoder.layers.14.self_attn.v_proj	1,328,256	12,004,448,688
model.vision_tower.encoder.layers.15.layer_norm1	2,304	12,004,450,992
model.vision_tower.encoder.layers.15.layer_norm2	2,304	12,004,453,296
model.vision_tower.encoder.layers.15.mlp.fc1	4,962,512	12,009,415,808
model.vision_tower.encoder.layers.15.mlp.fc2	4,959,360	12,014,375,168
model.vision_tower.encoder.layers.15.self_attn.k_proj	1,328,256	12,015,703,424
model.vision_tower.encoder.layers.15.self_attn.out_proj	1,328,256	12,017,031,680
model.vision_tower.encoder.layers.15.self_attn.q_proj	1,328,256	12,018,359,936
model.vision_tower.encoder.layers.15.self_attn.v_proj	1,328,256	12,019,688,192
model.vision_tower.encoder.layers.16.layer_norm1	2,304	12,019,690,496
model.vision_tower.encoder.layers.16.layer_norm2	2,304	12,019,692,800
model.vision_tower.encoder.layers.16.mlp.fc1	4,962,512	12,024,655,312
model.vision_tower.encoder.layers.16.mlp.fc2	4,959,360	12,029,614,672
model.vision_tower.encoder.layers.16.self_attn.k_proj	1,328,256	12,030,942,928
model.vision_tower.encoder.layers.16.self_attn.out_proj	1,328,256	12,032,271,184
model.vision_tower.encoder.layers.16.self_attn.q_proj	1,328,256	12,033,599,440
model.vision_tower.encoder.layers.16.self_attn.v_proj	1,328,256	12,034,927,696
model.vision_tower.encoder.layers.17.layer_norm1	2,304	12,034,930,000
model.vision_tower.encoder.layers.17.layer_norm2	2,304	12,034,932,304
model.vision_tower.encoder.layers.17.mlp.fc1	4,962,512	12,039,894,816
model.vision_tower.encoder.layers.17.mlp.fc2	4,959,360	12,044,854,176
model.vision_tower.encoder.layers.17.self_attn.k_proj	1,328,256	12,046,182,432
model.vision_tower.encoder.layers.17.self_attn.out_proj	1,328,256	12,047,510,688
model.vision_tower.encoder.layers.17.self_attn.q_proj	1,328,256	12,048,838,944
model.vision_tower.encoder.layers.17.self_attn.v_proj	1,328,256	12,050,167,200
model.vision_tower.encoder.layers.18.layer_norm1	2,304	12,050,169,504
model.vision_tower.encoder.layers.18.layer_norm2	2,304	12,050,171,808
model.vision_tower.encoder.layers.18.mlp.fc1	4,962,512	12,055,134,320
model.vision_tower.encoder.layers.18.mlp.fc2	4,959,360	12,060,093,680
model.vision_tower.encoder.layers.18.self_attn.k_proj	1,328,256	12,061,421,936
model.vision_tower.encoder.layers.18.self_attn.out_proj	1,328,256	12,062,750,192
model.vision_tower.encoder.layers.18.self_attn.q_proj	1,328,256	12,064,078,448
model.vision_tower.encoder.layers.18.self_attn.v_proj	1,328,256	12,065,406,704
model.vision_tower.encoder.layers.19.layer_norm1	2,304	12,065,409,008
model.vision_tower.encoder.layers.19.layer_norm2	2,304	12,065,411,312
model.vision_tower.encoder.layers.19.mlp.fc1	4,962,512	12,070,373,824
model.vision_tower.encoder.layers.19.mlp.fc2	4,959,360	12,075,333,184
model.vision_tower.encoder.layers.19.self_attn.k_proj	1,328,256	12,076,661,440

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Module	Params	Running total
model.vision_tower.encoder.layers.19.self_attn.out_proj	1,328,256	12,077,989,696
model.vision_tower.encoder.layers.19.self_attn.q_proj	1,328,256	12,079,317,952
model.vision_tower.encoder.layers.19.self_attn.v_proj	1,328,256	12,080,646,208
model.vision_tower.encoder.layers.20.layer_norm1	2,304	12,080,648,512
model.vision_tower.encoder.layers.20.layer_norm2	2,304	12,080,650,816
model.vision_tower.encoder.layers.20.mlp.fc1	4,962,512	12,085,613,328
model.vision_tower.encoder.layers.20.mlp.fc2	4,959,360	12,090,572,688
model.vision_tower.encoder.layers.20.self_attn.k_proj	1,328,256	12,091,900,944
model.vision_tower.encoder.layers.20.self_attn.out_proj	1,328,256	12,093,229,200
model.vision_tower.encoder.layers.20.self_attn.q_proj	1,328,256	12,094,557,456
model.vision_tower.encoder.layers.20.self_attn.v_proj	1,328,256	12,095,885,712
model.vision_tower.encoder.layers.21.layer_norm1	2,304	12,095,888,016
model.vision_tower.encoder.layers.21.layer_norm2	2,304	12,095,890,320
model.vision_tower.encoder.layers.21.mlp.fc1	4,962,512	12,100,852,832
model.vision_tower.encoder.layers.21.mlp.fc2	4,959,360	12,105,812,192
model.vision_tower.encoder.layers.21.self_attn.k_proj	1,328,256	12,107,140,448
model.vision_tower.encoder.layers.21.self_attn.out_proj	1,328,256	12,108,468,704
model.vision_tower.encoder.layers.21.self_attn.q_proj	1,328,256	12,109,796,960
model.vision_tower.encoder.layers.21.self_attn.v_proj	1,328,256	12,111,125,216
model.vision_tower.encoder.layers.22.layer_norm1	2,304	12,111,127,520
model.vision_tower.encoder.layers.22.layer_norm2	2,304	12,111,129,824
model.vision_tower.encoder.layers.22.mlp.fc1	4,962,512	12,116,092,336
model.vision_tower.encoder.layers.22.mlp.fc2	4,959,360	12,121,051,696
model.vision_tower.encoder.layers.22.self_attn.k_proj	1,328,256	12,122,379,952
model.vision_tower.encoder.layers.22.self_attn.out_proj	1,328,256	12,123,708,208
model.vision_tower.encoder.layers.22.self_attn.q_proj	1,328,256	12,125,036,464
model.vision_tower.encoder.layers.22.self_attn.v_proj	1,328,256	12,126,364,720
model.vision_tower.encoder.layers.23.layer_norm1	2,304	12,126,367,024
model.vision_tower.encoder.layers.23.layer_norm2	2,304	12,126,369,328
model.vision_tower.encoder.layers.23.mlp.fc1	4,962,512	12,131,331,840
model.vision_tower.encoder.layers.23.mlp.fc2	4,959,360	12,136,291,200
model.vision_tower.encoder.layers.23.self_attn.k_proj	1,328,256	12,137,619,456
model.vision_tower.encoder.layers.23.self_attn.out_proj	1,328,256	12,138,947,712
model.vision_tower.encoder.layers.23.self_attn.q_proj	1,328,256	12,140,275,968
model.vision_tower.encoder.layers.23.self_attn.v_proj	1,328,256	12,141,604,224
model.vision_tower.encoder.layers.24.layer_norm1	2,304	12,141,606,528
model.vision_tower.encoder.layers.24.layer_norm2	2,304	12,141,608,832
model.vision_tower.encoder.layers.24.mlp.fc1	4,962,512	12,146,571,344
model.vision_tower.encoder.layers.24.mlp.fc2	4,959,360	12,151,530,704

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Module	Params	Running total
model.vision_tower.encoder.layers.24.self_attn.k_proj	1,328,256	12,152,858,960
model.vision_tower.encoder.layers.24.self_attn.out_proj	1,328,256	12,154,187,216
model.vision_tower.encoder.layers.24.self_attn.q_proj	1,328,256	12,155,515,472
model.vision_tower.encoder.layers.24.self_attn.v_proj	1,328,256	12,156,843,728
model.vision_tower.encoder.layers.25.layer_norm1	2,304	12,156,846,032
model.vision_tower.encoder.layers.25.layer_norm2	2,304	12,156,848,336
model.vision_tower.encoder.layers.25.mlp.fc1	4,962,512	12,161,810,848
model.vision_tower.encoder.layers.25.mlp.fc2	4,959,360	12,166,770,208
model.vision_tower.encoder.layers.25.self_attn.k_proj	1,328,256	12,168,098,464
model.vision_tower.encoder.layers.25.self_attn.out_proj	1,328,256	12,169,426,720
model.vision_tower.encoder.layers.25.self_attn.q_proj	1,328,256	12,170,754,976
model.vision_tower.encoder.layers.25.self_attn.v_proj	1,328,256	12,172,083,232
model.vision_tower.encoder.layers.26.layer_norm1	2,304	12,172,085,536
model.vision_tower.encoder.layers.26.layer_norm2	2,304	12,172,087,840
model.vision_tower.encoder.layers.26.mlp.fc1	4,962,512	12,177,050,352
model.vision_tower.encoder.layers.26.mlp.fc2	4,959,360	12,182,009,712
model.vision_tower.encoder.layers.26.self_attn.k_proj	1,328,256	12,183,337,968
model.vision_tower.encoder.layers.26.self_attn.out_proj	1,328,256	12,184,666,224
model.vision_tower.encoder.layers.26.self_attn.q_proj	1,328,256	12,185,994,480
model.vision_tower.encoder.layers.26.self_attn.v_proj	1,328,256	12,187,322,736
model.vision_tower.post_layernorm	2,304	12,187,325,040